

DATA ANALYST PORTFOLIO

Stack Overflow Developer Survey Analytics: Technology Trends, Dashboard Insights & Demographic Analysis

*Transforming Raw Survey Data into Actionable Insights Through Data Wrangling,
Dashboarding, and Analytical Storytelling*

Prepared by:

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Tools & Technologies:

Excel | Power Query | SQL | Python | Jupyter | Cognos Analytics

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1. Summary

Built a multi-dashboard analytics solution from survey data by cleaning multi-select technology fields, normalizing the dataset, modelling relationships in Cognos Analytics, and producing insights on developer tool adoption, future preferences, and demographics.

2. Business / Analytical Question

What technologies are developers using today, what do they want to use next year, and how do those patterns vary within the broader respondent demographic profile?

3. Dataset

- Main dataset: Stack Overflow Developer Survey responses
- Supporting artefacts included:
 - job-postings.xlsx outputs
 - popular-languages.csv outputs
 - dashboard screenshots and analytics results compiled into [Data Analyst Capstone Project Report](#)

4. Tools Used

- Excel
- Power Query
- Python
- Jupyter Notebooks
- SQL/Relational Databases
- Cognos Analytics

5. Process (What I Did)

Data Preparation

- Identified survey fields containing semicolon-separated multi-select technology values
- Split the values into columns in Excel
- Unpivoted the split columns into rows using Power Query

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- Created **8 normalized CSV files** for separate technology dimensions
- Added the original survey table back for demographics

Data Modelling

- Built a Cognos data module
- Linked normalized technology tables and the original survey table using Responseld
- Structured the model to support dashboard analysis of languages, databases, platforms, web frameworks, and demographics.

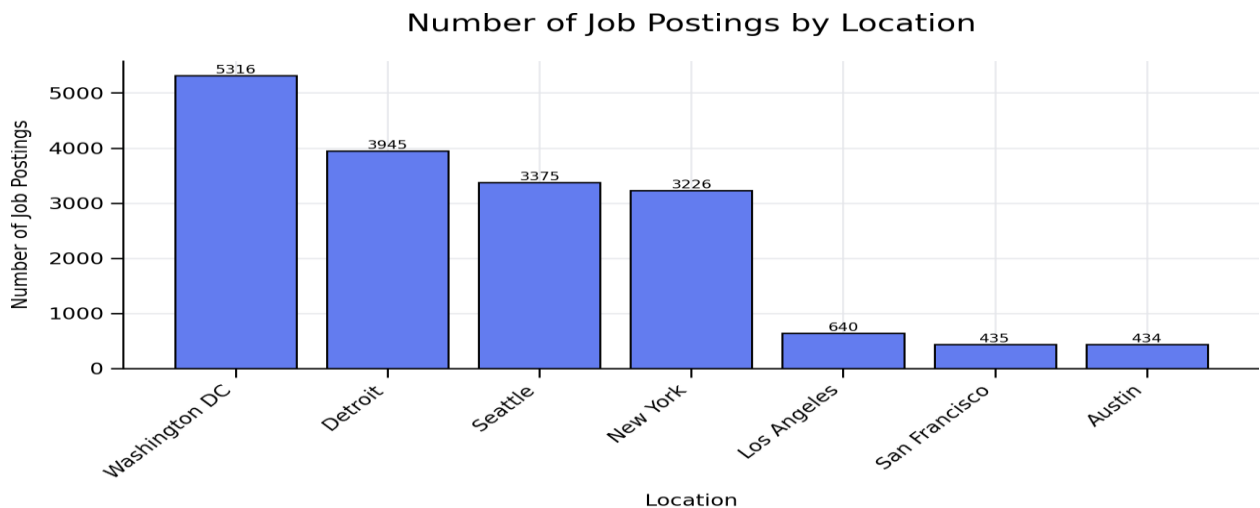
Dashboard Development

Created three dashboard tabs:

1. **Current Technology Usage**
2. **Future Technology Trend**
3. **Demographics**

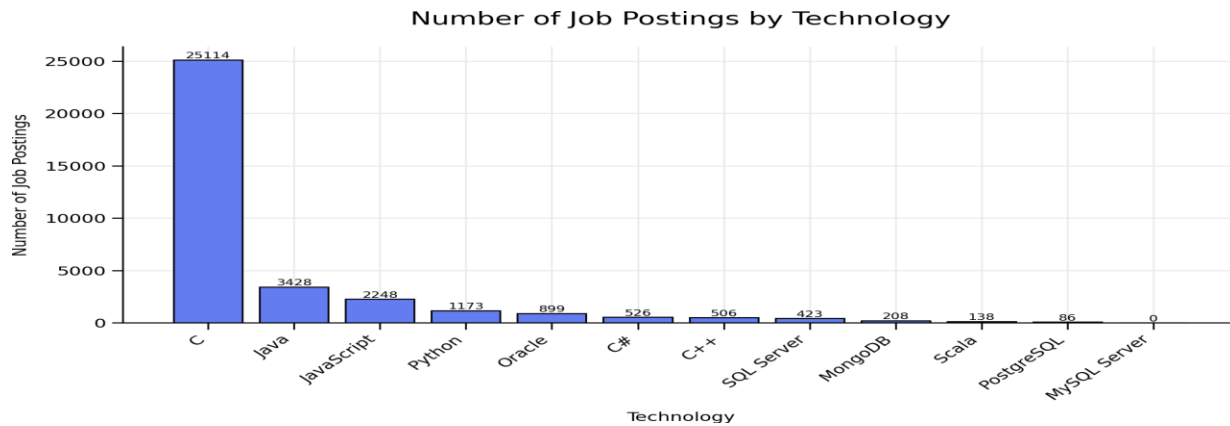
6. Key Visuals & Dashboard

(a) Job Postings by Location

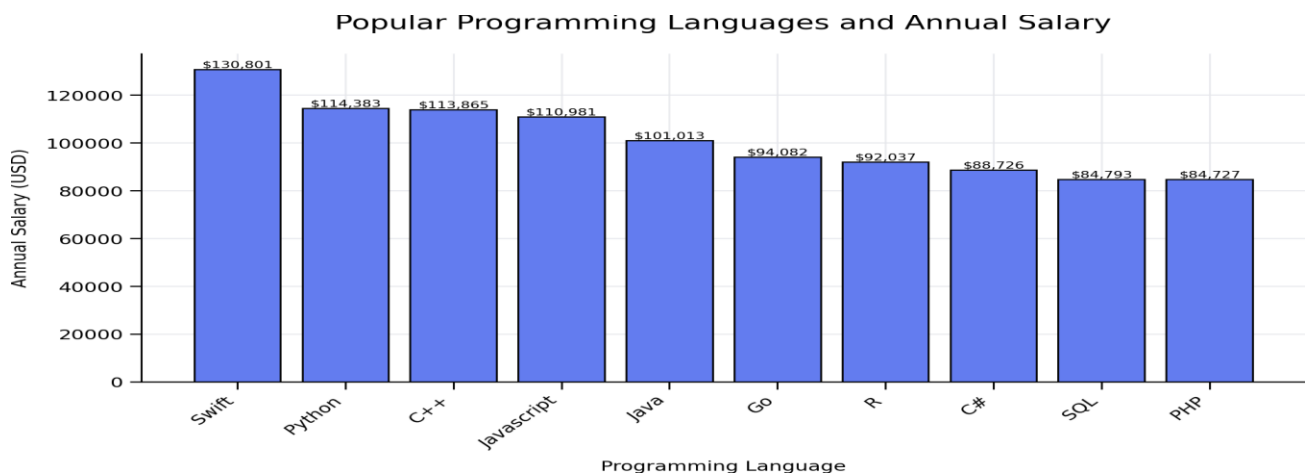


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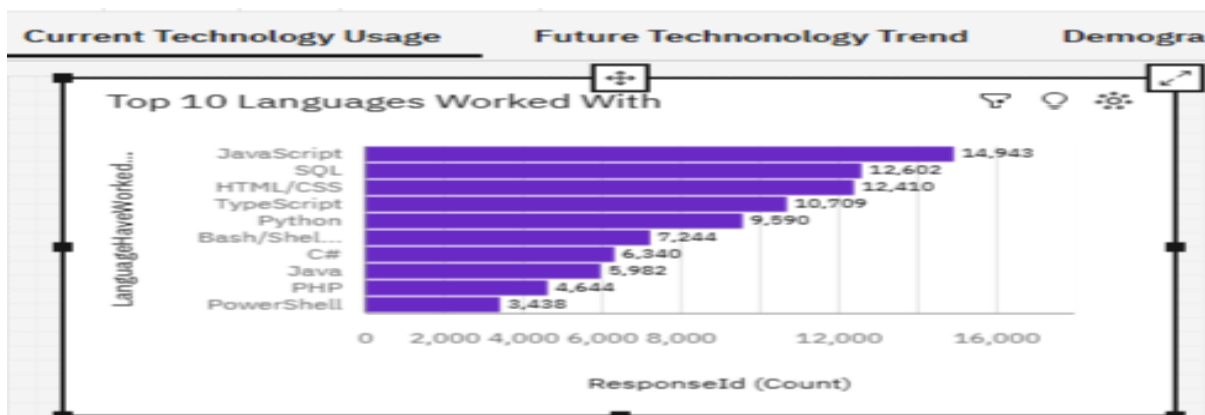
(b) Job Postings by Technology



(c) Popular Languages and Annual Salary

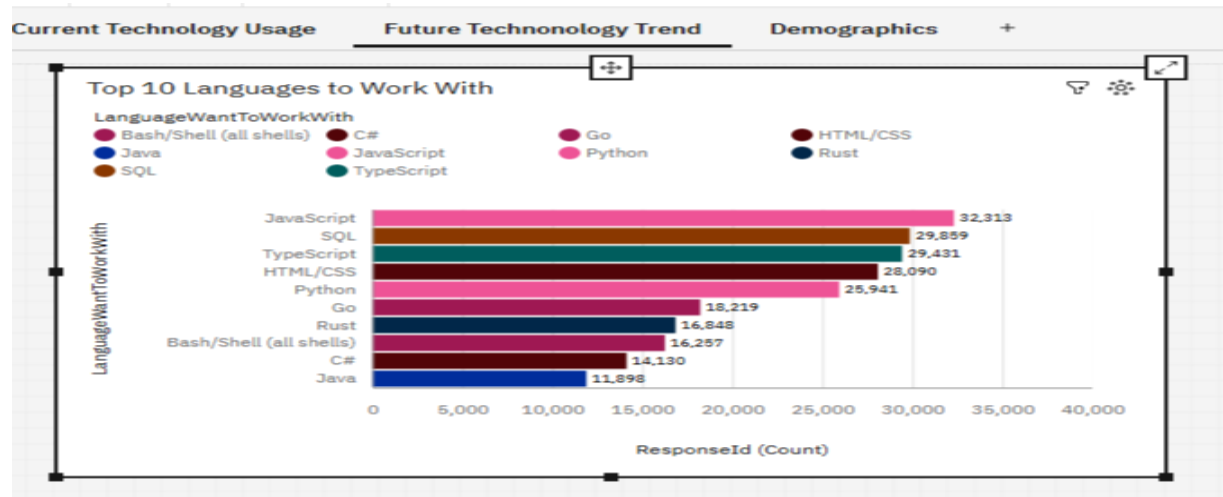


(d) Programming Language Trends Current Year



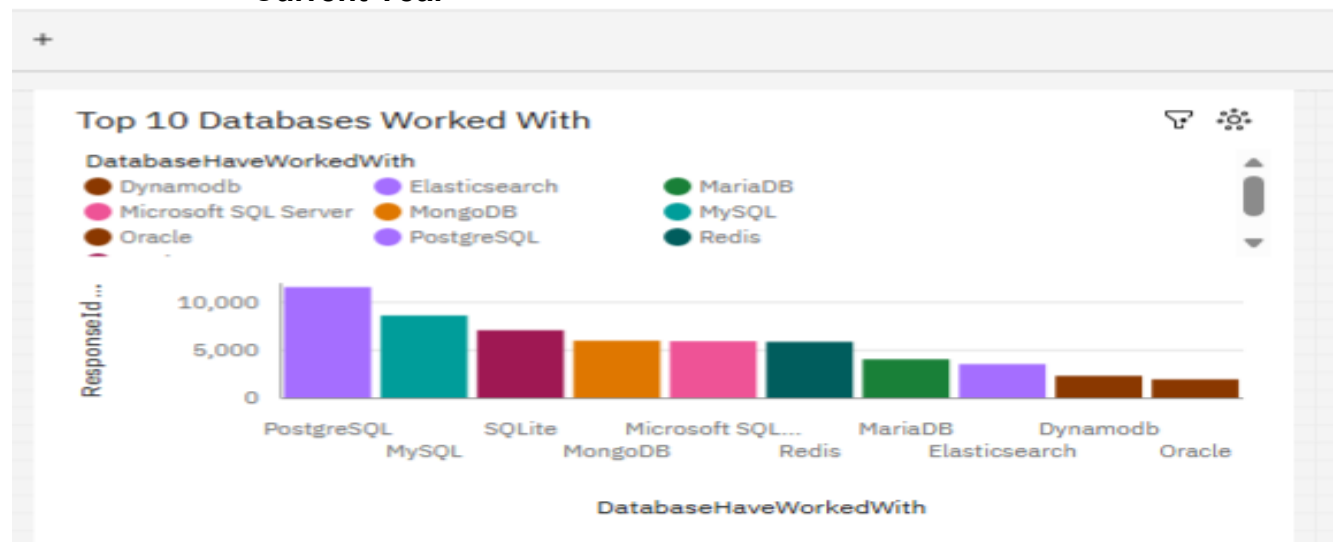
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Next Year



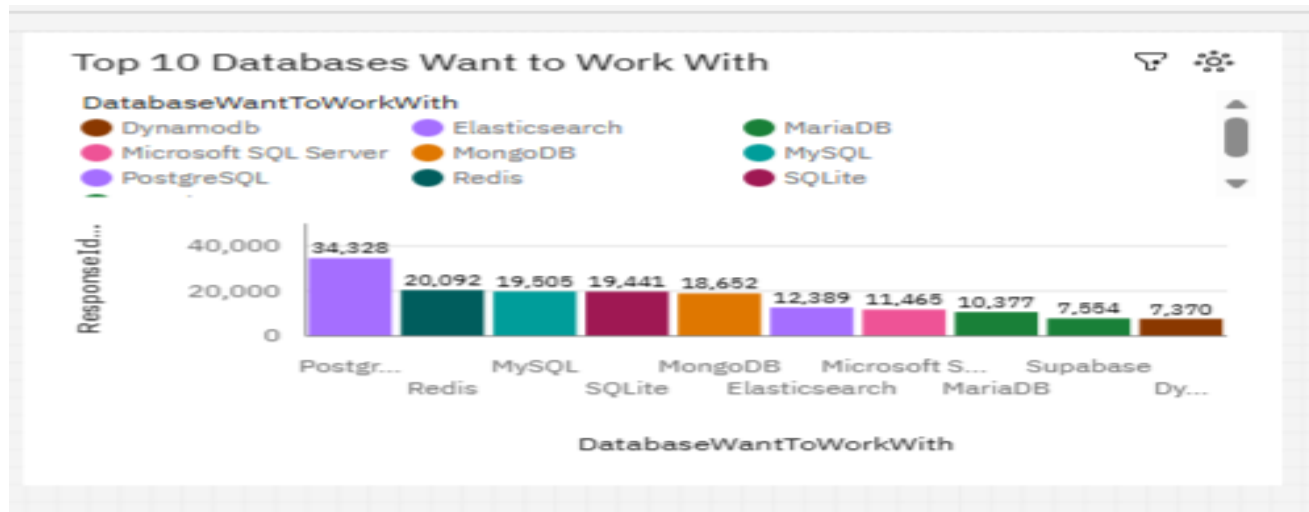
(e) Database Trends

Current Year



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Next Year



7. Key Insights

Current Technology Usage

- **JavaScript, SQL, HTML/CSS, and TypeScript** dominate current language usage
- **PostgreSQL** leads current database usage
- **AWS** leads current platform usage
- Mainstream cloud and web ecosystems dominate the current stack.

Future Technology Trend

- **JavaScript, TypeScript, and Python** remain dominant future-interest languages
- **PostgreSQL** remains the most desired database
- **AWS** remains the strongest desired future platform
- Future demand largely reinforces current adoption while increasing interest in **TypeScript, Rust, Supabase**, and more modern web tooling.

Demographics

- Largest respondent age group: **25–34 (41.3%)**
- Second largest: **35–44 (27.3%)**
- Third largest: **18–24 (15.9%)**

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- The strongest education profile is a **Bachelor's degree (8,629)** followed by **Master's degree (5,000)**.

8. Business Implications and Recommendations (Why This Project Matters)

Programming Languages Trends

- Organizations should continue to invest in JavaScript-based ecosystems
- Training, hiring, and curriculum planning should include stronger TypeScript capability
- Future-oriented teams may benefit from monitoring emerging demand for Rust

Database Trends

- Database strategy should continue to prioritise PostgreSQL-compatible talent and tooling
- Organizations should balance support for legacy relational databases with modern developer-first platforms
- Cloud-native and open-source database ecosystems are likely to remain influential.

9. Challenges and How to Solve Them

This project demonstrates that I can:

- Clean, messy real-world survey data,
- Transform multi-value fields into analysis-ready structures,
- Design data models for BI dashboards,
- Build interactive visual analytics in Cognos, and
- Turn raw data into decision-oriented insights.

10. Portfolio-Ready “What I’d Improve Next.”

To make the project even stronger in a production setting, I would:

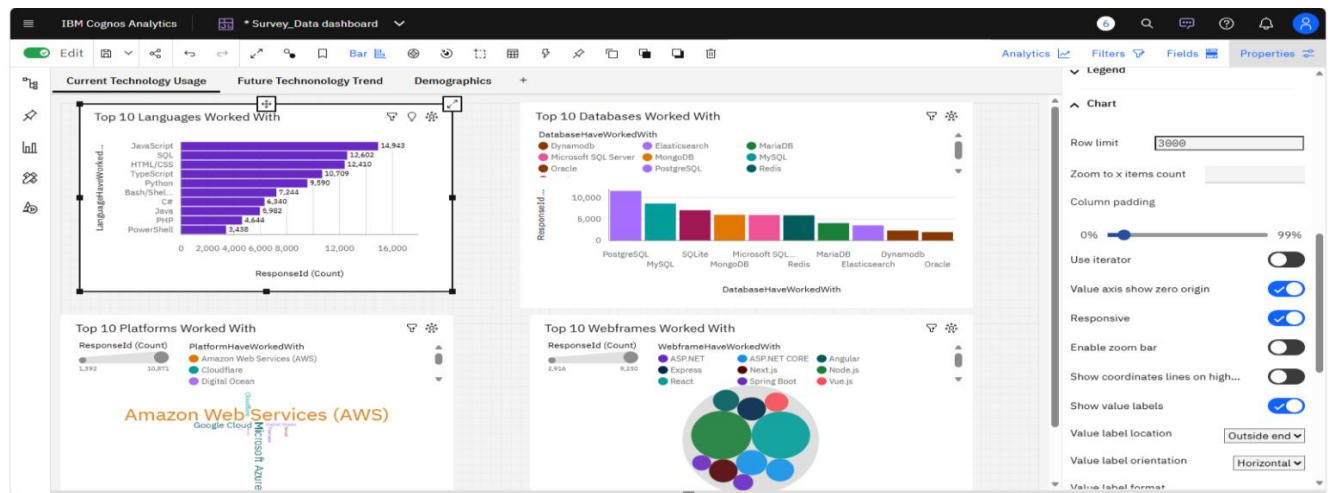
- automate more of the wrangling pipeline in Python,
- improve dashboard consistency and formatting,
- publish the dashboard narrative with a lighter, more executive design,
- and add a short recommendation memo for each dashboard audience.

11. Files and Links

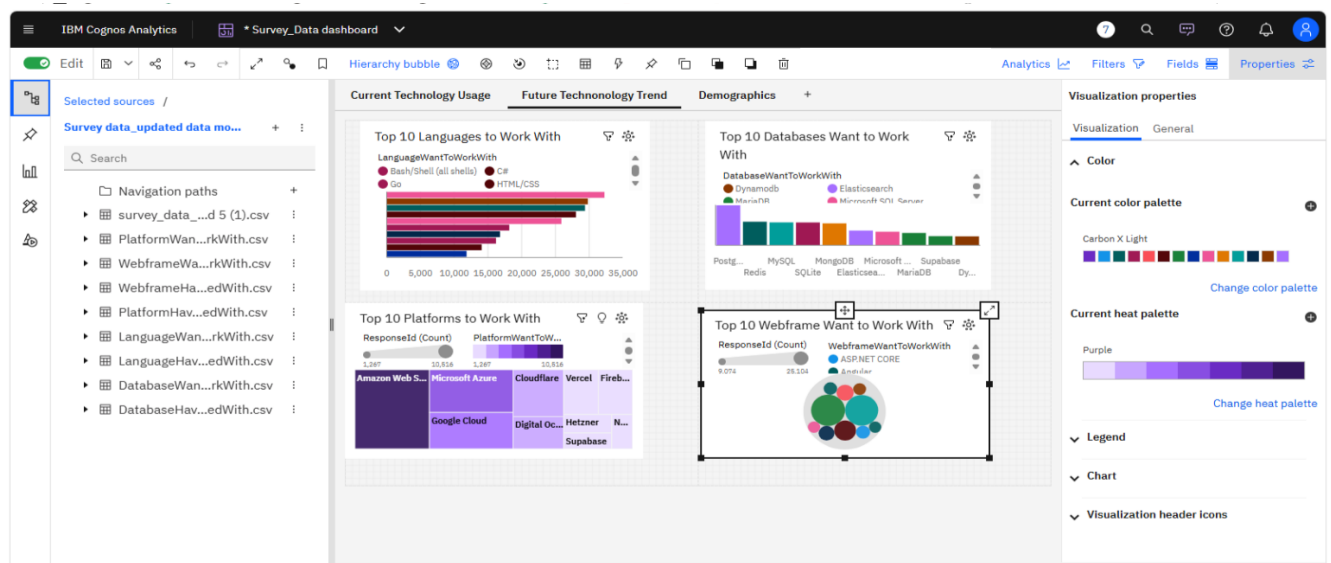
- notebook
- SQL script

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Dashboard Tab1: Current Technology Trends



Dashboard Tab2: Future Technology Trends



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Dashboard Tab3: Demographics

